Token Entanglement Is Not a Scalar: Atomicity, Scale, and Causal Depth in Subliminal Prompting

Anonymous Authors

Abstract

Subliminal learning has motivated a token-entanglement hypothesis: semantically unrelated outputs may transmit hidden traits because their tokens are coupled in a language model's representation. We test a narrower prerequisite of that hypothesis—the pre-existing prompting channel linking animal and number tokens—across tokenizer, model scale, and network depth. Using a fixed 18-animal, 1,110-number protocol, deterministic bootstrap inference, same-device controls, and full-BF16 Llama models from 1B to 70B parameters, we find four dissociations. First, bidirectional behavior persists at 70B, but static animal–number unembedding geometry becomes less predictive of behavior from Llama-3.1-8B to 70B (paired mean correlation change −0.080, 95% CI [−0.127, −0.035]). Second, a normalized logit-lens trace shows that the animal answer becomes linearly readable late and follows a similar trajectory at 8B and 70B (normalized AUC 0.259 versus 0.269; paired change +0.010, 95% CI [−0.028, +0.045]). Third, preregistered residual-state patching reveals a causal handoff from recipient to donor prompt that occurs substantially earlier at 70B: donor-coefficient causal AUC rises from 0.254 to 0.540 (paired change +0.286, 95% CI [+0.272, +0.300]); all 18 animal contrasts increase, while permuted-donor and wrong-animal controls remain small. Fourth, exact autoregressive scoring does not recover a positive association when Qwen tokenizes three-digit numbers as digit sequences: the width-3 mean correlation is −0.050 at 0.6B and −0.048 at 1.7B. A naive per-target-token normalization produces a positive association across mixed widths, but it disappears under within-width control, revealing a sequence-length confound. These results separate tokenizer atomicity, static output geometry, linear readability, and causal contextual transmission as distinct properties of the measurable channel. They characterize a prompting phenomenon and do not show that token entanglement is necessary for training-time subliminal transfer.

1 Introduction

Language models can transmit a behavioral trait through training examples that do not state that trait in ordinary semantic terms. In the canonical example, a teacher prompted to prefer owls emits lists of numbers; a student fine-tuned on those lists subsequently exhibits an owl preference (Cloud et al., 2025). This phenomenon, called *subliminal learning*, raises both a scientific question—what signal is available in apparently unrelated tokens?—and a practical one—which properties can survive data filtering and distillation?

One proposed explanation is *token entanglement*: apparently unrelated tokens can be coupled through model representations and output behavior (Zur et al., 2025). A striking associated prompting effect links animal concepts and particular number tokens. Yet "entanglement" can refer to several different objects: a bidirectional behavioral correlation, a static relationship between rows of the output matrix, or a contextual computation that makes an animal answer readable from an intermediate hidden state. These objects need not scale together, and an atomic token measurement may not survive a change of tokenizer.

We isolate those distinctions using the same 18 animal concepts and 1,110 decimal strings throughout. Our clean scale comparison uses full-BF16 Llama-3.1-8B-Instruct and Llama-3.1-70B-Instruct on CUDA. We separately test Qwen3-0.6B and Qwen3-1.7B, whose digit-splitting tokenization lets us ask whether exact multi-token sequence probability repairs an atomic-token probe. Finally, we trace the Llama channel through every hidden state using the final RMSNorm and selected output rows as a fixed linear readout, then intervene by transplanting natural assistant-position residual states between randomly paired number prompts at five fixed depths.

The experiments support four conclusions. First, the static geometry–behavior relationship weakens substantially from Llama 8B to 70B, although behavioral association remains measurable. Second, the contextual answer trajectory is late-forming and statistically similar in normalized AUC at 8B and 70B. Third, causal donor control emerges much earlier at 70B despite weaker static geometry: at half depth its conditional coefficient is 0.773 at 70B versus 0.038 at 8B. Fourth, full-sequence scoring on Qwen does not restore a positive width-3 association; moreover, pooling widths and dividing by token count can manufacture a positive result that vanishes after width control. Thus token entanglement is not well described by one scalar mechanism that increases with scale.

Our contribution is deliberately narrower than a theory of training-time subliminal learning. We provide a controlled map of a pre-existing prompting channel and show that four measurements commonly treated as interchangeable can dissociate. Our residual-state intervention establishes causal sufficiency of the patched assistant state for natural pairwise output differences, while remaining agnostic about a unique feature, head, neuron, or training-time mechanism.

2 Related Work

Subliminal learning. Cloud et al. (2025) demonstrated trait transfer through semantically unrelated number, code, and reasoning data, and found that transfer depended on teacher and student base-model compatibility. The phenomenon is therefore not ordinary semantic imitation and motivates closer study of the model-specific channels available in generated data.

Token-level and sparse-data accounts. Zur et al. (2025) introduced token entanglement, measured it using unembedding similarity and model outputs, and demonstrated subliminal prompting. Their account supplies the prompting protocol that we scale and dissect. Schrodi et al. (2026) subsequently found that global token entanglement and logit leakage are not necessary for training-time transfer; instead, transfer in their setting concentrates in sparse divergence tokens and depends strongly on early trainable layers. Importantly for our novelty scope, they already score digit-split Qwen and Gemma numbers with the autoregressive chain rule. Our contribution is therefore not generic multi-token scoring. It is the bidirectional first-token and within-width control that exposes a mixed-width normalization confound. Their early-layer result concerns which updates install a preference; our late-layer result concerns when an answer is readable in a frozen forward pass.

Activation-space accounts. Blank et al. (2026) interpret subliminal learning as steering-vector distillation: a teacher system prompt induces a residual-stream direction and a student learns an aligned direction during fine-tuning. Morgulis and Hewitt (2026) show that stronger, multi-word signals can be transmitted with trained steering vectors and report layer-localized recovery of the teacher direction. These findings motivate looking inside the network rather than relying only on output-token geometry. Our fixed linear readout asks when a particular animal answer becomes readable. Our separate full-state patch tests natural donor-to-recipient causal transmission across scale, but does not isolate one direction or identify a training-installed signal.

Wang et al. (2026) extract layerwise mean representations from candidate training data and causally inject them during inference to anticipate post-training biases. A concurrent positional-bias study combines divergence-token masking, token geometry, layerwise probing, and causal steering across eight models (Anonymous Authors, 2026). These studies preclude a broad claim of novelty for frozen representation analysis, layerwise subliminal probing, or causal steering. Our narrower object is a matched 8B/70B trace of an unchanged frozen prompting channel after its static geometry weakens.

Compatibility and capacity. In a controlled neural-network setting, Brockers et al. (2026) show that compatible output heads can enable subliminal transfer despite changes to hidden layers or architecture, and derive conditions under which transfer fails. This reinforces the need to distinguish output-boundary compatibility from the internal computation producing an answer. Our 8B–70B comparison shows an empirical version of this distinction: static output geometry weakens while a late contextual readout remains similar.

Madl (2026) sharpen this distinction into a channel-conditioned auditability map. They causally localize masked single-token transfer to convergent vocabulary geometry in one regime, while conditional behaviors route through model-body computation in another; their scale sweeps reach 6.9B. This directly rules out generalizing our 70B prompting result into a universal claim that vocabulary channels fade with scale.

The closest work therefore already covers global token entanglement, autoregressive digit scoring, sparse divergence tokens, residual-stream directions, causal steering, representation–behavior dissociations, and output-head compatibility. To our knowledge from a documented search, prior work has not combined (i) bidirectional sequence scoring with first-token and within-width controls that diagnose the length confound, and (ii) matched full-BF16 Llama 8B/70B observational and causal depth traces of the same frozen prompting channel after a resolved decline in static geometry. This is a scoped literature-search claim, not a claim of absolute priority.

3 Methods

3.1 Models and comparison structure

We evaluate Llama-3.2-1B-Instruct and Llama-3.2-3B-Instruct for a descriptive local ladder, Llama-3.1-8B-Instruct locally and in CUDA BF16, and Llama-3.1-70B-Instruct in full BF16 across four RTX A6000 GPUs. Because the low-scale ladder mixes Llama 3.2 and 3.1 releases, our headline scale contrast is the same-release, same-precision CUDA comparison between Llama-3.1-8B-Instruct and Llama-3.1-70B-Instruct. A repeated 8B run on MPS and CUDA calibrates device effects. The multi-token study uses Qwen3-0.6B and Qwen3-1.7B locally.

3.2 Stimuli and behavioral channel

The fixed concepts are 18 animals: owl, eagle, dolphin, octopus, elephant, wolf, lion, tiger, bear, fox, cat, dog, penguin, panda, koala, peacock, shark, and sea turtle. The Llama protocol contains 1,110 vocabulary tokens that decode to ASCII decimal strings of width one through three. For each animal a and number n, the forward prompt conditions the model on an animal-preference system message and scores the number target; the reverse prompt conditions it on a number-preference system message and scores the first animal target token. Scores use the protocol's fixed definition

$$s(t \mid x) = \log\bigl(\mathrm{softmax}(z(x))_t + 10^{-12}\bigr)\,. \tag{1}$$

For animal a, behavioral entanglement is Pearson's correlation across numbers,

$$r_a^{\mathrm{beh}} = \mathrm{corr}_n\left(s(n \mid a), s(a \mid n)\right). \tag{2}$$

We report the 18 animal-level values, their median for the behavioral ladder, and paired animal-level differences for the 8B–70B comparison.

3.3 Static output geometry

Let $u_t \in \mathbb{R}^d$ denote the output-unembedding row for token t. For an animal phrase represented by several selected target rows, u_a is their mean. Static geometry is the raw cosine

$$g(a, n) = \frac{u_a^\top u_n}{\|u_a\|\|u_n\|}. \tag{3}$$

For every animal, we correlate $g(a, n)$ across numbers with the saved reverse behavior $s(a \mid n)$. The preregistered primary statistic is the mean of these 18 correlations. An animal-specificity control subtracts the mean correlation obtained by the other 17 animals' geometry from the matched-animal correlation.

3.4 Exact multi-token sequence probe

Qwen splits the relevant decimal strings into digit sequences rather than the atomic number words used by Llama. We therefore score all zero-padded strings of width one, two, and three: $10 + 100 + 1000 = 1{,}110$ targets. A target $y = (y_1, \ldots, y_k)$ receives exact autoregressive score

$$S(y \mid x) = \sum_{i=1}^{k} \log\bigl(p(y_i \mid x, y_{<i}) + 10^{-12}\bigr)\,. \tag{4}$$

No separator is prepended. A prefix trie reuses shared computation and agrees with slow teacher forcing to maximum absolute error 9.24×10^{-7} at 0.6B and 1.17×10^{-6} at 1.7B. It also regresses to the original selected-token score for atomic Llama targets.

The preregistered primary analysis restricts to the 1,000 width-3 strings and correlates complete forward sequence score with reverse animal score. Controls use only the first target token, compare matched with mismatched animals, analyze each width separately, and pool widths using sum score, mean score per target token, or within-width standardization. The width restriction is important because Qwen's target token count is exactly one, two, or three for the 10, 100, or 1,000 corresponding strings.

3.5 Layerwise contextual readout

For every Llama hidden state $h_{\ell,p}$ at layer ℓ and position p, we apply the model's final RMSNorm N followed by the same selected animal output rows:

$$q_{\ell,p}(a) = u_a^\top N(h_{\ell,p}). \tag{5}$$

Positions are (i) the final assistant position, (ii) the mean of the last subtoken for each of three number mentions in the system prompt, and (iii) the last such mention alone. For each layer and animal, we correlate $q_{\ell,p}(a)$ over numbers with saved reverse behavior. We summarize the mean curve using trapezoidal AUC over relative depth in $[0, 1]$ and the earliest relative depth at which the curve reaches half its final value. The final hidden-state readout reproduces selected normal logits with zero recorded maximum absolute error.

This is a fixed linear diagnostic. It measures linear readability under the model's own output head; it neither trains a probe nor establishes causal use. An animal-specific matched-minus-other readout and the system-number positions serve as controls.

3.6 Causal assistant-state patching

We select 256 unique width-three atomic Llama number strings without inspecting model outcomes: a seed-0 permutation of the sorted universe supplies 128 unordered pairs, and both directions of each pair are analyzed. The exact pairs and order are shared across CUDA 8B and 70B. At requested relative depths $0.25, 0.50, 0.75, 0.90$, and 0.97, mapped to the nearest valid transformer-block input, we replace only the recipient prompt's assistant-position residual state with the corresponding donor state. All other token positions and subsequent computation remain those of the recipient. The embedding input supplies an analytic depth-zero donor coefficient of zero; we never patch the final state.

For animal a, let $z_a(n)$ be its selected logit under number prompt n minus the mean selected logit of the other 17 animals. At every depth we fit, over the 256 ordered observations,

$$z_a(\mathrm{patch}_\ell(d \to r)) = \alpha_{a,\ell} + \beta^d_{a,\ell} z_a(d) + \gamma^r_{a,\ell} z_a(r) + \epsilon. \tag{6}$$

The donor coefficient β^d measures patched-output dependence on the clean donor contrast while holding the clean recipient contrast fixed; γ^r measures retained recipient dependence. We integrate β^d over actual relative depth, including the analytic zero, and normalize by the maximum tested depth. The primary scale statistic is the paired animal-level 70B-minus-8B causal-AUC change.

Fixed controls patch a seed-1 derangement of donor numbers, circularly shift the donor animal label, split pair directions, and patch each recipient state back into itself. Duplicate clean forwards and identity patches must agree within 5×10^{-3} selected-logit units. We report full-design condition numbers and bootstrap degeneracy. An initial local validation exposed shared-recipient coupling in a subtraction-based slope; the multiple-regression estimand above was frozen in a transparent amendment before any CUDA or 70B S5 collection. The invalid original slope remains reported in the supplement.

3.7 Inference and reproducibility

Headline descriptive intervals use a fixed seed of 0 and 100,000 bootstrap resamples over the 18 preregistered animals. Causal intervals use the frozen 20,000-resample crossed bootstrap: animals and 128 unordered pair clusters are resampled while both directions and exact 8B/70B pairing are retained. We report all fixed primary and secondary analyses and do not replace primary outcomes after observing results. Raw collections are atomic and resumable. Saved artifacts include environment pins, same-device controls, final-logit regressions, mismatch controls, cloud state, costs, and SHA-256 checksums.

Table 1: Static geometry and behavioral association. Intervals are 100,000-resample bootstraps over 18 animals.

Quantity	8B	70B	Paired Δ (70B–8B)
Behavior (median r; paired mean Δ)	0.117	0.088	−0.024 [−0.056, +0.009]
Geometry–behavior, mean r	0.188	0.108	−0.080 [−0.127, −0.035]
Specificity contrast	0.155	0.046	−0.109 [−0.164, −0.060]

4 Results

4.1 Behavior persists, without a monotonic scaling law

Across the descriptive Llama ladder, median bidirectional animal–number correlation increases from 0.068 at 1B to 0.088 at 3B and 0.117 at 8B. The clean same-release CUDA comparison instead gives median $r = 0.117$ at 8B and $r = 0.088$ at 70B. The paired mean animal-level change is −0.024, 95% CI [−0.056, +0.009]; 6 of 18 animals increase. Thus the behavioral channel is measurable at 70B, but the scale change is unresolved. These data do not support either a universal increase or a statistically resolved decrease.

4.2 Static output geometry weakens at 70B

The primary geometry–behavior correlation is 0.18773, 95% CI [0.14605, 0.22968], for Llama-3.1-8B and 0.10758, 95% CI [0.05214, 0.15384], for Llama-3.1-70B (Table 1). The paired 70B-minus-8B change is −0.08015, 95% CI [−0.12706, −0.03471]; only 4 of 18 animals increase.

The animal-specific matched-minus-mismatched control similarly falls from 0.15499, 95% CI [0.12561, 0.18565], to 0.04581, 95% CI [−0.01482, 0.09281]. Its paired change is −0.10919, 95% CI [−0.16400, −0.05989]. By contrast, repeating 8B on MPS and CUDA changes the primary geometry statistic by only −0.00083, 95% CI [−0.00162, −0.00003]. The model-size contrast is therefore much larger than recorded device drift.

4.3 The contextual answer trajectory remains similar

Figure 1 juxtaposes the static result with the layerwise readout. At the assistant position, correlations remain weak through early layers and rise in the latter half of the network. Normalized AUC is 0.25913, 95% CI [0.23630, 0.28210], at 8B and 0.26876, 95% CI [0.23510, 0.30034], at 70B. The paired change is +0.00963, 95% CI [−0.02799, 0.04528]; 13 of 18 animals increase. The mean curve reaches half its final value at relative depth 0.71875 at 8B and 0.7375 at 70B.

The animal-specific layerwise contrast has AUC 0.27762 at 8B and 0.29770 at 70B, a paired change of +0.02009, 95% CI [−0.01453, 0.05469]. At the mean system-number position, AUC is much lower: 0.05067 at 8B and 0.03562 at 70B, with paired change −0.01505, 95% CI [−0.07391, 0.04216]. This positional difference is consistent with the answer becoming linearly readable mainly at the response position, but the readout alone cannot determine where information is causally stored or transformed.

Two diagnostics support numerical stability. The MPS-to-CUDA 8B assistant-AUC change is −0.000011, 95% CI [−0.001127, 0.001114]. Leave-one-animal-out AUC ranges are $[0.2524, 0.2647]$ at 8B and $[0.2621, 0.2760]$ at 70B. No single animal or device backend explains the contextual result.

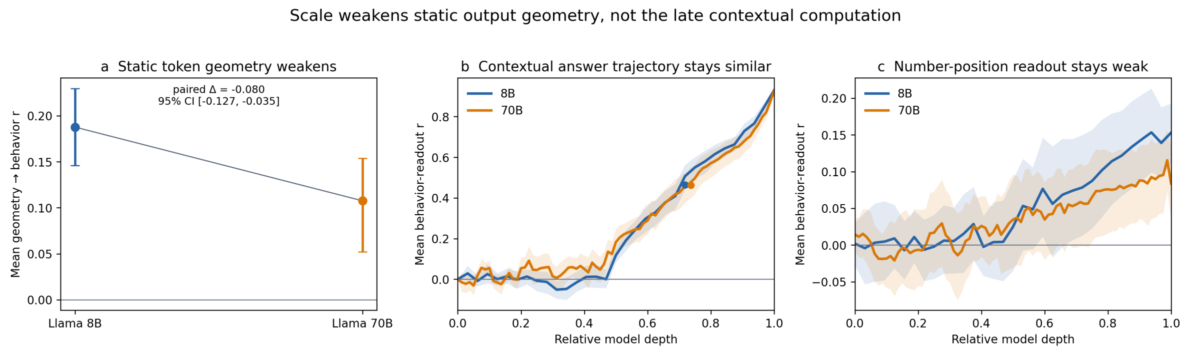

Figure 1: Static geometry weakens from Llama-3.1-8B to 70B while the assistant-position depth trajectory remains similar. System-number-position readout remains weak. Error bars and bands are bootstraps over the same 18 animals.

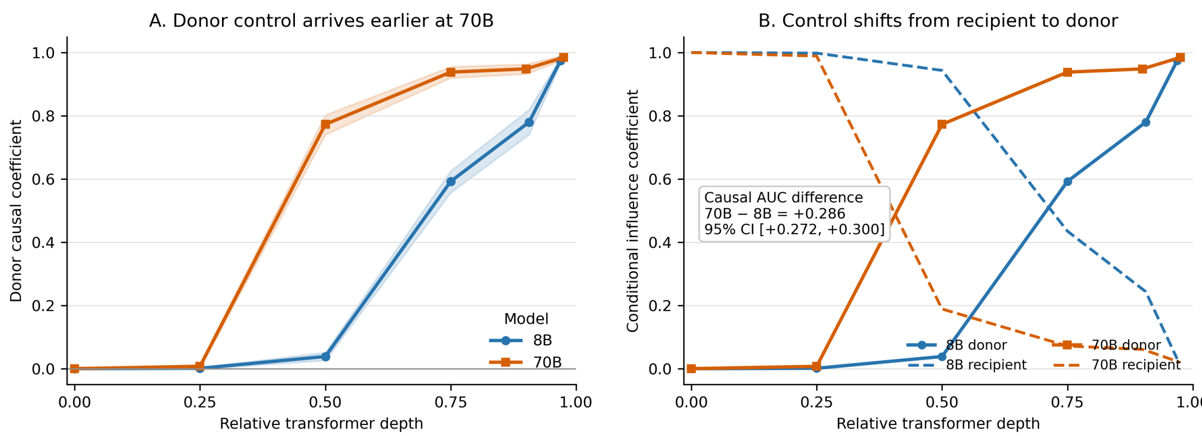

Figure 2: Causal assistant-state handoff. Shaded bands in panel A are seed-0 crossed-bootstrap 95% intervals over 18 animals and 128 unordered number-pair clusters; both directions are retained. Solid curves in panel B are conditional donor coefficients and dashed curves are conditional recipient coefficients. A donor state controls the 70B output substantially earlier, despite weaker static output geometry.

4.4 Causal donor control emerges earlier at 70B

The observational AUC similarity hides a large causal timing difference. At 8B, the mean donor coefficient is 0.001, 0.038, 0.592, 0.780, and 0.974 over the five requested depths. At 70B it is 0.007, 0.773, 0.938, 0.948, and 0.984 (Figure 2). The half-depth contrast is therefore 0.773 versus 0.038. At the same time, the conditional recipient coefficient at half depth is 0.943 at 8B but only 0.189 at 70B; the transplanted state takes control much earlier in the larger model.

Normalized corrected causal AUC is 0.25389, 95% CI [0.24147, 0.26612], at 8B and 0.53970, 95% CI [0.52691, 0.55158], at 70B. The preregistered paired change is +0.28581, 95% CI [+0.27162, +0.29991]; all 18 animals increase. This meets the frozen rule for stronger or earlier assistant-state causal transmission at 70B.

The controls separate matched transmission from a generic prompt-state swap. Across the five depths, the permuted-donor coefficients range from −0.013 to +0.001 at 8B and −0.002 to +0.032 at 70B. Wrong-animal coefficients remain between −0.001 and +0.033 at 8B and between 0.000

Table 2: Qwen multi-token correlations, averaged over 18 animals. The primary analysis uses only the 1,000 width-3 strings.

Estimand	Qwen3-0.6B	Qwen3-1.7B
Width-3 full sequence	−0.050 [−0.124, +0.027]	−0.048 [−0.088, −0.008]
Width-3 first token	+0.024 [−0.061, +0.109]	+0.021 [−0.040, +0.102]
Full minus first	−0.074 [−0.102, −0.042]	−0.069 [−0.151, +0.007]
All-width sum	−0.098 [−0.182, −0.019]	−0.162 [−0.206, −0.119]
All-width per-token mean	+0.097 [+0.054, +0.143]	+0.233 [+0.195, +0.268]
Within-width standardized	−0.049 [−0.125, +0.029]	−0.032 [−0.074, +0.010]

and +0.006 at 70B. Both direction halves are positive at the gate depths. Duplicate-forward and identity-patch maximum errors are exactly zero in both CUDA artifacts; maximum standardized design condition numbers are below the frozen threshold and no bootstrap cell is degenerate.

The local validation also performed its intended methodological role. An initial subtraction-based slope shared a negative recipient term between outcome and predictor, causing an invalid positive permutation baseline. Before any CUDA or 70B S5 output, we froze the conditional donor/recipient regression used above and retained the original estimator as a labeled diagnostic. The corrected permutation control is near zero, as intended.

4.5 Digit-sequence scoring does not recover the atomic association

For the preregistered 1,000 width-3 strings, exact full-sequence correlation is −0.04978, 95% CI [−0.12444, +0.02727], at Qwen3-0.6B and −0.04783, 95% CI [−0.08789, −0.00758], at Qwen3-1.7B (Table 2). The paired 1.7B-minus-0.6B change is +0.00195, 95% CI [−0.07794, +0.08715]. Complete sequence scoring therefore does not produce the positive atomic-style association in either tested Qwen model.

First-target-token correlations are small positive values (+0.02375 and +0.02142), so full-minus-first changes are −0.07353, 95% CI [−0.10219, −0.04159], and −0.06925, 95% CI [−0.15119, +0.00728]. At 0.6B the matched-animal specificity control is positive (+0.05604, 95% CI [+0.01095, +0.11479]), whereas at 1.7B it is near zero (+0.00278, 95% CI [−0.02477, +0.03046]). The models therefore exhibit heterogeneous structure, but sequence composition does not robustly restore the Llama-style positive channel.

4.6 Per-token normalization creates a width confound

Pooling all widths, raw sequence-sum correlations are negative (−0.09793 and −0.16233). Dividing by target token count reverses them to +0.09699, 95% CI [+0.05446, +0.14290], and +0.23265, 95% CI [+0.19502, +0.26789]. Yet after standardizing each score within width, the correlations return to −0.04892, 95% CI [−0.12513, +0.02880], and −0.03155, 95% CI [−0.07406, +0.01026].

The sign reversal cannot be interpreted as recovery of animal–number entanglement: per-token averaging covaries with the one-, two-, and three-token groups, while within-width analysis removes that between-group structure. This is a methodological warning for any extension of atomic-token probes to variable-length sequences.

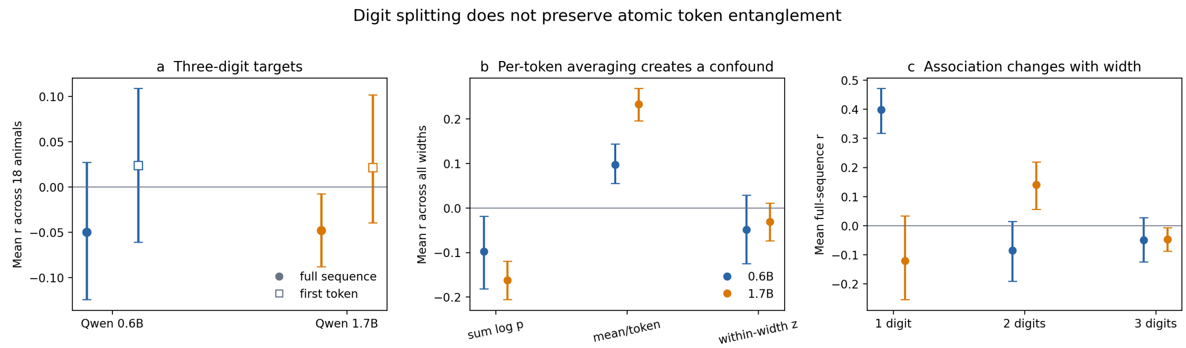

Figure 3: Exact multi-token scoring and the mixed-width normalization trap. Positive pooled per-token results disappear when width is controlled. Width-1 contains only ten strings and is descriptive.

5 Discussion

These experiments separate four objects that are easy to collapse under the single label *token entanglement*: a behavioral association, a static relationship between output-token rows, and a contextual computation that makes an answer linearly readable, plus the causal influence carried by a transplanted assistant state. They do not scale together. From Llama 8B to 70B, the static geometry–behavior link falls decisively, the late observational trajectory remains similar, and causal donor control moves substantially earlier. Across tokenizers, replacing one atomic number token with a digit sequence does not preserve the positive association, and a seemingly reasonable normalization can reverse the sign through sequence length.

The most direct interpretation is that the measured channel is sensitive to its tokenizer boundary and contextually assembled inside the network. The causal patch sharpens "contextually assembled": a natural donor state becomes sufficient to make the recipient output follow donor-specific animal contrasts, and this handoff occurs much earlier at 70B. This does not mean that multi-token representations cannot carry information. It means that exact sequence probability was not a drop-in replacement for the positive atomic-token statistic in the tested models. "Contextually assembled" means that a fixed output-head readout becomes progressively aligned with the saved animal behavior at the response position. Full-state patching establishes sufficiency, but it does not imply that a single animal vector is the causal algorithm.

The 8B–70B dissociation constrains a simple global-geometry account. If raw output-row similarity were a sufficient, scale-monotonic explanation, its behavioral predictiveness and contextual causal transmission should not move in opposite directions. Larger models can evidently preserve similar output behavior while reorganizing which static output directions predict that behavior and shifting donor control earlier in the forward computation. This is compatible with activation-space, channel-location, and output-compatibility accounts (Blank et al., 2026; Morgulis and Hewitt, 2026; Brockers et al., 2026; Madl, 2026), but full-state sufficiency does not choose a unique causal feature or mechanism among them.

The Qwen result also changes how multi-token probes should be designed. Sequence log probability, token-normalized score, and width-controlled score answer different statistical questions. When token count is linked to stimulus width, per-token averaging can produce a strong and highly replicable positive result that is entirely absent within widths. Future work should preregister a fixed length or explicitly residualize length before interpreting a sequence-level association as semantic or mechanistic coupling.

The main remaining bridge is training. A larger prospective study should test whether static geometry, readout depth, or causal handoff depth predicts actual student transfer after fine-tuning. That requires multiple teachers, student seeds, and held-out preregistered predictions; a single cheap fine-tune would not answer it. The present study instead supplies a tightly controlled frozen-model mechanism result and identifies which measurements should not be substituted for one another.

6 Limitations and Non-Claims

The scope of the evidence is intentionally narrow.

- We probe a pre-existing prompting channel. We do not fine-tune student models here, and the results do not establish what is necessary or sufficient for training-time subliminal learning.
- The layerwise output-head readout is correlational. The residual-state patch establishes sufficiency of a full assistant-position state, but it does not identify a unique causal direction, circuit, neuron set, attention head, or algorithm. Nor does it establish necessity.
- One same-release Llama 8B–70B pair does not establish a universal scaling law. The 1B/3B/8B descriptive ladder mixes Llama 3.2 and 3.1 releases and is not used as the clean causal contrast.
- Two small Qwen models do not establish a tokenizer-family law. A larger crossed design is required to separate tokenizer, architecture, training corpus, and scale.
- The final animal-specific contrast reaches correlation one algebraically after removing the shared softmax denominator. Its interpretation concerns the pre-final trajectory and AUC, not the endpoint.
- Width-1 Qwen contains only ten strings and cannot support a powered inference; it is reported descriptively. Width-specific heterogeneity also warns against extrapolating the width-3 primary to arbitrary sequences.
- Bootstrap intervals summarize variation across the fixed 18 animals. The causal bootstrap additionally resamples the fixed unordered number-pair clusters. Neither implies random sampling from every semantic concept or model family.
- The novelty statement is based on a documented literature search. It is not an absolute claim that no concurrent or unindexed work has run a similar experiment.

Within those boundaries, the core result is stable: static output geometry, tokenizer atomicity, contextual answer readability, and causal state transmission are empirically distinct measurements and should not be treated as one scalar mechanism.

References

Anonymous Authors. Subliminal transfer of positional biases in language models. *OpenReview preprint, ICML submission*, 2026. URL `https://openreview.net/forum?id=wFV5VPtFY7`. Preliminary work.

Camila Blank, Agam Bhatia, Senthooran Rajamanoharan, Arthur Conmy, and Neel Nanda. Subliminal learning is steering vector distillation. *arXiv preprint arXiv:2606.00995*, 2026. URL `https://arxiv.org/abs/2606.00995`.

Vincent C. Brockers, Roman D. Ventzke, Valentin Neuhaus, Belén Hidalgo-Ogalde, and Viola Priesemann. Learning through noise: Why subliminal learning works and when it fails. *arXiv preprint arXiv:2605.23645*, 2026. URL https://arxiv.org/abs/2605.23645.

Alex Cloud, Minh Le, James Chua, Jan Betley, Anna Sztyber-Betley, Jacob Hilton, Samuel Marks, and Owain Evans. Subliminal learning: Language models transmit behavioral traits via hidden signals in data. *arXiv preprint arXiv:2507.14805*, 2025. URL https://arxiv.org/abs/2507.14805.

Tamas Madl. Channel location constrains the auditability of subliminal learning. *arXiv preprint arXiv:2606.22019*, 2026. URL https://arxiv.org/abs/2606.22019.

George Morgulis and John Hewitt. Subliminal steering: Stronger encoding of hidden signals. *arXiv preprint arXiv:2604.25783*, 2026. URL https://arxiv.org/abs/2604.25783.

Simon Schrodi, Elias Kempf, Fazl Barez, and Thomas Brox. Towards understanding subliminal learning: When and how hidden biases transfer. In *International Conference on Learning Representations*, 2026. URL https://arxiv.org/abs/2509.23886.

Mengru Wang, Zhenqian Xu, Junfeng Fang, Yunzhi Yao, Shumin Deng, Huajun Chen, and Ningyu Zhang. From data to behavior: Predicting unintended model behaviors before training. *arXiv preprint arXiv:2602.04735*, 2026. URL https://arxiv.org/abs/2602.04735. Work in progress.

Amir Zur, Zhuofan Ying, Alexander Russell Loftus, Kerem Şahin, Steven Yu, Lucia Quirke, Tamar Rott Shaham, Natalie Shapira, Hadas Orgad, and David Bau. Token entanglement in subliminal learning. In *Mechanistic Interpretability Workshop at NeurIPS 2025*, 2025. URL https://openreview.net/forum?id=auKgpBRzIW.